

Module Summary

Project 1: AI Programming Foundations

Overview

In this project i built a small but complete reproducible data workflow that combines two real climate datasets and walks them through download, cleaning, exploratory analysis, visualization and summary in a Jupyter notebook. The two datasets are the **Berkeley Earth Land + Ocean monthly temperature record** (1850 - present, <https://berkeleyearth.org/data/>) and the **NOAA Mauna Loa monthly atmospheric CO₂ record** (1958 - present, <https://gml.noaa.gov/ccgg/trends/>). The deliverable is the workflow itself, no machine learning model is trained at this stage (Danchev, 2022).

Dataset Description

The Berkeley Earth file reports a global monthly **temperature anomaly in degrees Celsius**, going back to 1850 and expressed relative to the Jan 1951 - Dec 1980 average, together with uncertainty estimates and 1, 5, 10 and 20-year centred running means (methodology in Rohde & Hausfather, 2020; data file Rohde & Hausfather, 2025). After removing the file's two stacked alternative reconstructions the working frame is 2,100 monthly rows × 12 numeric columns. The Mauna Loa file is a clean monthly CSV from NOAA's Global Monitoring Laboratory with 816 rows × 8 columns (Lan et al., 2026); the key variables are *average* (raw monthly CO₂ in ppm) and *deseasonalized* (NOAA's seasonally-adjusted trend). The two records are merged on a mid-month Date key with an inner join, producing 802 monthly overlap rows from March 1958 to December 2024 for the joint analysis.

Workflow Description

The notebook runs as a sequence of small, independent steps so each can be re-run, replaced or extended on its own (Danchev, 2022):

- **Ingestion:** Both files are pulled directly from their official URLs into a local *data/* folder via an idempotent download function driven by a small *DATASETS* dict.
- **Cleaning:** Two functions, *clean_temperature* and *clean_co2*, add a mid-month Date column, reorder + sort, and (for CO₂) replace NOAA's negative sentinel codes with NaN. A third small step inner-joins the cleaned frames.
- **Exploratory analysis:** *.describe()* on both clean frames, a reusable *summarize_by_decade(df, value_col)* function used twice, the Pearson correlation in the merged frame, and the five hottest/coldest months on record.
- **Visualizations:** Three figures (long-term temperature with running mean and uncertainty band, the Keeling curve with raw and deseasonalized trends, and an annual-mean CO₂-vs-temperature scatter with linear fit), each with title, labelled axes and a written interpretation.
- **Summary:** A written summary in the notebook ties the three figures back to the science, with a short subsection on planned extensions for later projects.

Key Decisions and Assumptions

The most consequential cleaning step is **converting NOAA's sentinel missing-value codes** (-1, -9.99, -0.99) into proper NaN. Without it pandas would silently average those negative codes into any aggregation and quietly produce wrong numbers, explicit and consistent representation of missing data is a long-standing best practice (Danchev, 2022). I deliberately chose **not to drop early temperature rows** even though they contain NaNs in the running-mean columns: those NaNs are expected (a centred 10-year mean cannot exist for the first/last 5 years) and the full 175 years of history is the whole reason the dataset was chosen.

I treated **temperature as anomalies, not absolute values**, because absolute global means are extremely sensitive to which stations are in the network at a given time while anomalies cancel out station-specific biases, this is the standard convention used by Berkeley Earth, HadCRUT, NASA GISTEMP and the IPCC (Rohde & Hausfather, 2020; IPCC, 2021). Every plot axis is labelled with the baseline so a reader cannot misread +1 as an absolute temperature.

For EDA I aggregated to **decades** to smooth single-year noise (volcanic eruptions, El Niño / La Niña years) without losing the long-term shape, and aggregated to **annual means** specifically for Figure 3, because monthly data is too noisy to read a clean correlation off a scatter. Each figure was designed to answer one question: Figure 1, “how has global temperature changed long-term?”; Figure 2, “how has CO₂ changed since 1958?”; Figure 3, “how do the two relate?”. Figure 1 also marks the conventional start years of the 2nd, 3rd and 4th industrial revolutions (Schwab, 2016) as faint vertical lines so the reader can see the trend in historical context.

Results and Interpretation

The three figures together support a single clear story. From 1850 until roughly the late 1970s the 10-year mean global temperature anomaly stays in a narrow band a little below the 1951 - 1980 baseline (Figure 1). From the late 1970s onward it climbs steeply, the most recent years sit well above +1 °C, and atmospheric CO₂ has risen from about 315 ppm in 1958 to over 420 ppm today with no slowdown visible in the deseasonalized trend (Figure 2). Aggregated to annual means, the Pearson correlation between Mauna Loa CO₂ and the temperature anomaly reaches $r \approx 0.97$ for the 1958 - 2024 overlap, and the trajectory through time walks almost straight up and to the right (Figure 3).

The fact that the climb only really begins around 1970, even though CO₂ had already been rising for decades, has a well-documented two-part explanation. Sulfate aerosols from uncontrolled coal and oil burning between roughly 1945 and 1970 reflected sunlight back to space and partially cancelled out CO₂ warming, the period of “global dimming” documented in Wild (2009). Once clean-air laws kicked in (US Clean Air Act 1970, similar laws across Europe), aerosol concentrations dropped fast and the underlying CO₂ signal became visible. The IPCC AR6 (2021) attributes essentially all warming since 1970 to human activity with very high confidence.

Responsible Practice (Bias and Data Quality)

Several aspects of the data and the cleaning choices could mislead a reader if not handled explicitly:

- **Anomaly vs absolute confusion:** Treating +1.5 on Figure 1 as “1.5 °C absolute” instead of “1.5 °C above the 1951 - 1980 baseline” is the most common misreading in climate communication. Mitigated by labelling every axis with the baseline and stating the absolute value (≈ 14.10 °C) in the dataset description.
- **The 1940s warming bump:** The visible bump around 1940 in Figure 1 is mostly a measurement artifact, US ships using engine-intake thermometers temporarily replaced UK ships using cooler bucket-based sea-surface measurements during WWII, biasing SSTs warm (Thompson et al., 2008). Berkeley Earth’s corrections reduce but do not fully remove this discontinuity.
- **19th-century coverage is sparse:** The uncertainty band on Figure 1 is visibly wider before 1900 because far fewer stations were reporting; trends drawn from that era alone should be treated cautiously.
- **CO₂ from a single point:** Mauna Loa is a global proxy because the well-mixed free troposphere is representative, but it is not a global average in the same sense the temperature record is. The seven-month substitute period at Maunakea (Nov 2022 - Jul 2023) is marked on Figure 2.
- **Aggregation hides regional variation:** Both records are global means, so they cannot speak to regional disparities (e.g. Arctic warming faster than the tropics) or distributional impacts. Adding ocean-only temperature, regional breakdowns or paleo CO₂ records is the natural next step.

Reproducibility

Another person can rerun the entire workflow from a fresh clone with three commands: clone the repo, `pip install -r requirements.txt`, then run all cells in `data_workflow.ipynb`. Dependencies are pinned with exact versions in `requirements.txt`, generated from the working environment using `pip freeze` and pruned to the direct dependencies; pinning the computational environment is one of the explicit recommendations in Danchev (2022). The notebook is idempotent, the download cell skips files that already exist locally, and the cleaning, EDA and visualization cells produce the same outputs from the same inputs every time. The Git workflow uses an initial commit capturing the complete project on `main` plus a separate `development` branch with at least one further commit, which is merged back into `main` before submission, so the repository carries multiple commits and a branch beyond `main` as required. The `.gitignore` excludes the local virtual environment, downloaded data files (which the notebook regenerates on demand) and Jupyter checkpoint folders, so the repository contains only what another person needs to reproduce the work.

Sources and Citations

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education*, 5(56), 156. <https://doi.org/10.21105/jose.00156>

IPCC. (2021). *Climate change 2021: The physical science basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (V. Masson-Delmotte et al., Eds.). Cambridge University Press. <https://doi.org/10.1017/9781009157896>

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